Journal of Hardware and Systems Security (2026) 10:12
https://doi.org/10.1007/s41635-026-00182-4

AI-Assisted Side-Channel Analysis for Golden-Chip Free Detection and Localization of Hardware Trojans

Ashutosh Ghimire[1] · Tanvir Hossain[2] · Fathi Amsaad[1] · Tamzidul Hoque[2]

Received: 2 September 2025 / Accepted: 20 May 2026
© The Author(s) 2026

Abstract

The increasing threats to semiconductor integrated circuits (ICs) from hardware Trojans create growing risks of data breaches, system malfunctions or even device destruction. Traditional detection methods use Golden IC comparisons together with destructive reverse engineering techniques. The research introduces a hybrid framework which unites on-chip localization with unsupervised machine learning clustering eliminating need of golden data for Trojan detection. The on-chip localization technique reduces the reverse-engineering scope from full-chip inspection to a smaller RO-neighborhood region associated with the highest Trojan-impact score. The validation process on Field Programmable Gate Arrays (FPGAs) demonstrates that the detection system achieves 95% accuracy for small, short-triggered Trojans across various Trojan sizes and designs. Unlike existing golden-free, the main contribution of this work consists of uniting localization with clustering to enhance IC security approaches.

Keywords Hardware Trojan · FPGA · Side-channel analysis · Localization

1 Introduction

Research in hardware Trojan detection and prevention has rapidly evolved in recent years to maintain trust in integrated circuits (ICs) design, fabrication, and validation stages. Unfortunately, the IC manufacturing process is being increasingly outsourced to offshore facilities to reduce design costs, improve time-to-market, and meet the growing demand for IC in embedded computing applications [1]. Relying on overseas fabrication facilities presents substantial trust concerns, as individuals within untrusted facilities may introduce malevolent alterations or hardware Trojans, thereby triggering functional failures, compromising the security of sensitive data, or causing reliability issues [2, 3].

Many approaches have been proposed to detect hardware Trojans at various phases of the IC supply chain, with a particular emphasis on detection within untrusted foundries. Among different post-silicon detection methods, invasive techniques such as reverse engineering (RE) based Hardware Trojan (HT) detection approaches are accurate and reliable; however, they are not feasible for testing a broad population of ICs since these approaches are exceedingly time-consuming, costly, and damaging due to the chip destruction that occurs following RE. The destructive RE process requires decapsulation, de-processing, and advanced imaging equipment, making it impractical for widespread deployment.

The non-destructive hardware security approaches include side-channel analysis (SCA) that provides faster detection capabilities, rendering them well-suited for assessing large quantities of ICs [4]. SCA involves the analysis of one or more parametric behaviors, including current, voltage, electromagnetic radiation, and temperature of

Ashutosh Ghimire and Tanvir Hossain contributed equally to this work.

✉ Ashutosh Ghimire
ashutosh.ghimire@wright.edu

✉ Fathi Amsaad
fathi.amsaad@wright.edu

Tanvir Hossain
tanvir@ku.edu

Tamzidul Hoque
hoque@ku.edu

[1] Dept. of Computer Sci. and Eng., Wright State University, Colonel Glenn Hwy, Dayton, OH 45435, USA

[2] Department of Electrical Eng. and Computer Sci., University of Kansas, W 15th St, Lawrence, KS 66045, USA

Published online: 24 June 2026

the suspect IC, to detect the presence of Trojans. Existing studies have also introduced on-chip sensors like ring oscillators (ROs) to facilitate more precise SCA-based detection [5]. However, these SCA techniques are susceptible to environmental and manufacturing process variations. Since process variability in nanoscale technologies increases, it is hard to determine if a deviation is due to hardware Trojan or process-induced variation. Therefore, addressing variability-related issues has received much attention in recent research on SCA.

Furthermore, the application of supervised machine learning (ML) can enhance this approach by enabling it to learn the differentiation between the side-channel characteristics of chips free from Trojans and those containing Trojans [6]. Nonetheless, a notable limitation of SCA-based detection lies in the necessity of having golden ICs, which are typically assumed to be available through destructive RE. Golden data from multiple trusted ICs is required to address process variations, which significantly limits scalability.

Several methods are proposed to prevent the necessity for golden IC for side channel hardware security detection. Several security techniques that apply self-referencing use SC data of the same suspect component obtained at a different time or from the different identical location of the suspect chip [4, 7, 8]. These methods remove the requirement for golden ICs and eliminate the difficulty of inter-die and intra-die process changes, making side-channel analysis-based detection more useful. However, spatial self-referencing depends on the exact location (e.g., cores of multi-core processor) with at least one location as Trojan-free, and temporal self-referencing cannot detect combinational Trojans [8].

ML is proposed for golden-IC-free detection techniques. The work presented in [9] applies a supervised learning-based approach to identify anomalies in supply current induced by Trojans, utilizing simulation-based modeling. Nevertheless, it's worth noting that this study's training and test datasets are derived from simulations. Hence, the effectiveness of classifying data from fabricated chips has yet to be discovered. Other works, such as [10], explore brain-inspired or temporal learning models to reduce reliance on golden data.

When HT gets triggered, abrupt changes happen in the side-channel traces. Like the human brain, HTM can detect the anomaly occurring in the system. More recently, unsupervised clustering methods are proposed that eliminate the need for a golden chip for SCA but perform RE of a few ICs if an anomaly is detected. For instance, [11] divides all tested ICs into clusters by analyzing their power signatures. Since ICs with and without Trojans are likely to form different clusters, the integrity of all chips in a cluster can be verified by performing RE on only one IC from each cluster. Therefore, this approach reduces the required reverse engineering of ICs to the number of clusters formed, unlike traditional golden-chip-based approaches where a large number of RE would be needed to create the golden signature.

A similar approach has been explored in [12] using quantum diamond microscope magnetic field images,. However, both [12] and [11] suggest full RE of only a few ICs, which still require significant time, cost, and expertise. The present work builds upon these efforts by advancing golden-free detection with the integration of unsupervised clustering and a novel on-chip localization algorithm. This combination not only eliminates the reliance on golden ICs but also provides spatial information to narrow down the reverse engineering region of interest. Although no benchmark TrustHub Trojans are included in the current experiments, the framework demonstrates up to 95% detection accuracy on FPGA prototypes, offering practical insights into scalability for future ASIC evaluation.

1.1 Threat Model

We consider the foundry to be untrusted where an adversary has physical access to the hardware layout, masks, and HDL files to perform malicious modification. We do not consider Trojans that do not add or remove logic. The scope of this research does not encompass analog circuit-based Trojans, Trojans involving doping-level modifications, or other forms of non-digital attack vectors [13]. The RO network used for detection is assumed to be inserted by the trusted design house before fabrication as part of the design-for-test and security-monitoring infrastructure. We do not assume that the RO network is hidden from the foundry. Instead, the RO network acts as a distributed sensing fabric whose expected existence, connectivity, and readout behavior are known to the design house. Therefore, if an untrusted foundry removes, disables, or significantly tampers with the RO network, the design house can detect such tampering during post-silicon test because the expected RO responses or readout paths would be missing or inconsistent. The adversary is therefore allowed to know the presence of the RO sensors, but cannot modify both the functional logic and the RO monitoring infrastructure without creating detectable inconsistencies in the collected RO measurements.

1.2 Contributions of the Current Paper

This work introduces a golden-free framework that combines unsupervised ML clustering with on-chip localization for hardware Trojan detection. Ring oscillator (RO) networks are distributed across the IC during design, and after

fabrication their frequency data are analyzed through clustering. Unlike prior golden-free clustering approaches [11, 12], the framework additionally provides spatial localization of likely Trojan impact, enabling reverse engineering to focus on a specific region instead of the full die. This reduces verification effort while increasing detection confidence.

The main contributions are summarized as follows:

- A golden-free detection framework is presented that integrates unsupervised clustering with on-chip RO measurements, avoiding the need for destructive reverse engineering to obtain golden signatures.
- A localization algorithm is introduced that estimates the Trojan's likely on-chip region from frequency variations, thereby reducing the scope of required reverse engineering. The localization result is reported as a ranked RO-neighborhood estimate, where the highest-scoring ROs indicate the most likely physical region affected by Trojan-induced power-grid disturbance. To the best of our knowledge, this capability has not been reported in prior golden-free approaches.
- The framework is evaluated on FPGA prototypes of an MD5 processor core with combinational and sequential Trojan insertions of multiple sizes.
- Five clustering models (K-means, AGNES, BIRCH, GMM, and HMM) are comparatively assessed. While HMM achieves the highest accuracy (95%), the contribution lies in the comparative analysis of multiple clustering models for golden-free detection rather than proposing HMM as a novel technique itself.
- The study acknowledges limitations, including evaluation on FPGA rather than ASIC and the absence of benchmarks. Despite these, t he experiments demonstrate consistent detection performance across models and the results provide new insights into the feasibility of combining clustering with on-chip localization for practical golden-free detection.

1.3 Organization of Paper

The paper follows a structured format with subsequent sections adhering to a specific organizational scheme. Section 2 provides a comprehensive overview of previous research, an explanation of the threat model, and the presentation of essential background information. Section 3 is dedicated to a detailed description of the experimental setup employed in the proposed method, while Section 4 elaborates on the methodology proposed for Trojan detection. Experimental results are presented in Section 5, and Section 6 is dedicated to comprehensively discussing these findings. The paper concludes in Section 7.

2 Background

2.1 Hardware Trojan

Hardware Trojans are one of the popular circuitry modifications with malicious intent and the ability to elude standard detection methods. It is intentionally altered to cause harm or achieve a malicious objective, posing a significant threat to electronic systems. The Trojan comprises two key components: the payload, which is responsible for modifying circuit behavior upon activation of the trigger and initiating hardware Trojan under specific conditions. This model is visually represented in Fig. 1 [14]. Both the trigger and payload induce measurable side-channel impacts, and the proposed framework does not attempt to distinguish between the two explicitly; instead, deviations at the RO level capture their combined effects. Hardware Trojans are classified based on their payload, activation attributes, and design properties, highlighting their diverse characteristics. Understanding and addressing these traits are essential for effective countermeasures against this pervasive threat in electronic systems.

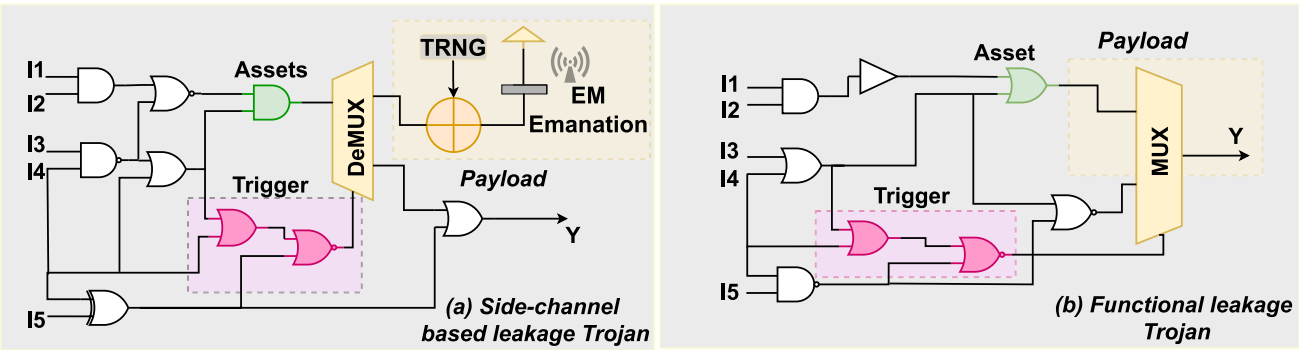


Fig. 1 Hardware Trojan to leak sensitive data through the leakage. (**a**) leak sensitive data through side-channel, (**b**) leak sensitive data through functional outputs

2.2 Unsupervised Machine Learning

ML is a research domain dedicated to learning from a collection of exemplar data, often called training data, to make predictions or classifications for new, unseen data, commonly known as testing data. It is conventionally categorized into two main branches: supervised learning and unsupervised learning, contingent upon the presence or absence of class labels in the training data. Many research studies predominantly rely on supervised learning, assuming all training data is accurately labeled as Trojan-free or Trojan-inserted. In realistic scenarios, such labels are unavailable because the exact type or location of Trojan insertion is unknown. Instead, unsupervised learning becomes essential, since it operates without labels during training. Clustering is a prominent technique within unsupervised learning, involving grouping data samples into clusters based on shared characteristics. In this work, labels are used only during evaluation to compute accuracy, F1 score, and related metrics, and not for training.

2.3 Data Sample Collections and Analysis

A RO is an electronic circuit consisting of an odd number of serially connected inverters interconnected in a closed loop configuration. as depicted in Fig. 2. The oscillation frequency of the RO is determined by a combination of the delay introduced by the amplifiers within the loop and the propagation delay stemming from the interconnecting wires. Its frequency is highly sensitive to supply voltage fluctuations and process variations, which makes it suitable for side-channel monitoring [15].

Alternative sensors, such as the carry4 chain in FPGA fabrics, have also been explored in prior work because of their sensitivity to circuit changes [16]. However, the present study focuses on ROs due to their well-established role in side-channel Trojan detection and their suitability for frequency-based clustering [5, 17].

The mathematical expression for the RO frequency (*f*) of an N_stage RO is formally defined by Eq. 1.

$$f = \frac{\mu_g \times (V_{DD} - V_{threshold})^{\alpha}}{2N \times k} \tag{1}$$

where the supply and threshold voltages are V_{DD} and $V_{threshold}$, respectively. Also, α represents the velocity saturation index, while the parameter k is a constant value dependent on the gate configuration. Furthermore, the μ_g defined the carrier mobility of the device and can be expressed by Eq. 2.

$$\mu = \frac{e}{4 \cdot \pi \cdot \hbar^2 \cdot p_s \cdot k_B \cdot T} \cdot \int_0^{\infty} \frac{E \cdot \tau(E) dE}{\cos h\left(\frac{E-\zeta}{2 \cdot k_B \cdot T}\right)} \tag{2}$$

Here, in this Eq. 2, if we consider other terms as constant, it shows that mobility depends on the Temperature (T). The electron mobility is also susceptible to the process variation (V), channel length (L), load capacitance (C_L), and threshold (V_{th}). New research on using RO networks to detect malicious post-silicon attacks is proposed. Because RO-based protection is shown to protect integrated circuits (ICs) against two distinct forms of attack: (a) inclusion attacks and (b) removal attacks. Given the current trend towards highly compact and power-constrained IC designs, malicious alteration or inclusion within the circuit can induce a significant voltage drop and additional noise. Because Trojan insertion typically increases switching activity or induces IR-drop on the power grid, it impacts the RO frequency. Thus, both combinational and sequential Trojans leave detectable side-channel signatures. If the attacker includes any malicious artifacts in the post-silicon process, the chip frequency will

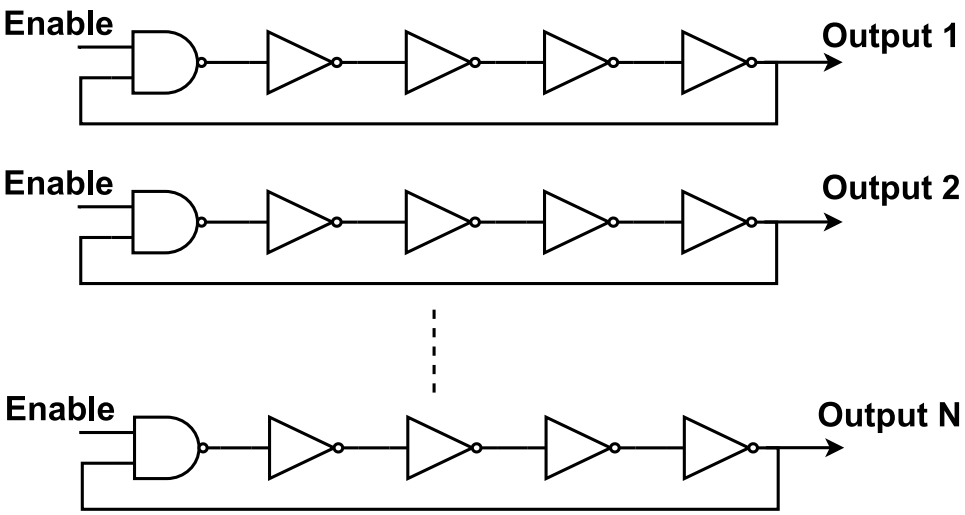


Fig. 2 Schematic of RO Array

be affected. Thus, the design house can detect the alteration while testing the chip. Also, if the attacker removes any circuit or RO network itself, the design house can still see the circuit removal. Because the design house will try to test the chip with ROs and, hence, will not be able to get valid data if any removal occurred. In this regard, RO networks have emerged as a promising solution for mitigating the impact of such attacks on ICs (Fig. 3).

2.4 Related Works

The detection of foundry-inserted Trojans is predominantly approached through two widely discussed techniques: side-channel analysis and logic testing, as described in [18]. Logic testing requires the Trojan's circuity to be fully activated (not partially), which poses a significant challenge in generating test vectors capable of activating all potential Trojans. Furthermore, logic testing is limited to detecting Trojans that impact the IC's functionality, and therefore, it cannot detect side-channel leakage Trojans, as illustrated in Fig. 1 [19].

In contrast to traditional testing methods, side-channel analysis is a unique approach for uncovering Trojans in ICs. This technique involves scrutinizing a suspect IC's physical traits, such as electromagnetic radiation, heat generation, transient current, leakage current, delay, or energy consumption, to identify the potential presence of a Trojan [20]. Notably, the benefits of SC analysis lie in its ability to detect Trojans without requiring full activation. Nevertheless, the perforce of SC approaches is challenged by both process manufacturing variations and environmental factors, as emphasized by [21].

Most SCA-based methods typically rely on a set of golden or Trojan-free ICs to establish a reference side-channel signature. Recently, techniques are proposed that do not require golden chips, utilizing electromagnetic and light emission parameters of suspect devices to create a model of the golden signature through simulation and trusted netlist and layout analysis [22, 23]. However, discrepancies between simulation-derived models and silicon may require more reliable detection methods. Additionally, observing light emission requires backside repackaging [24], and these techniques are primarily evaluated using Trojan-free ICs.

Researchers have explored applying ML techniques to eliminate the need for golden ICs. In [9], a supervised learning-based detection method using supply current data is proposed, but the training data are obtained through simulation, and the technique still needs to be tested on supply current data from fabricated chips

Unsupervised clustering has been applied to power signatures [11] and magnetic field images [12]. These methods reduce the reverse-engineering burden but still require

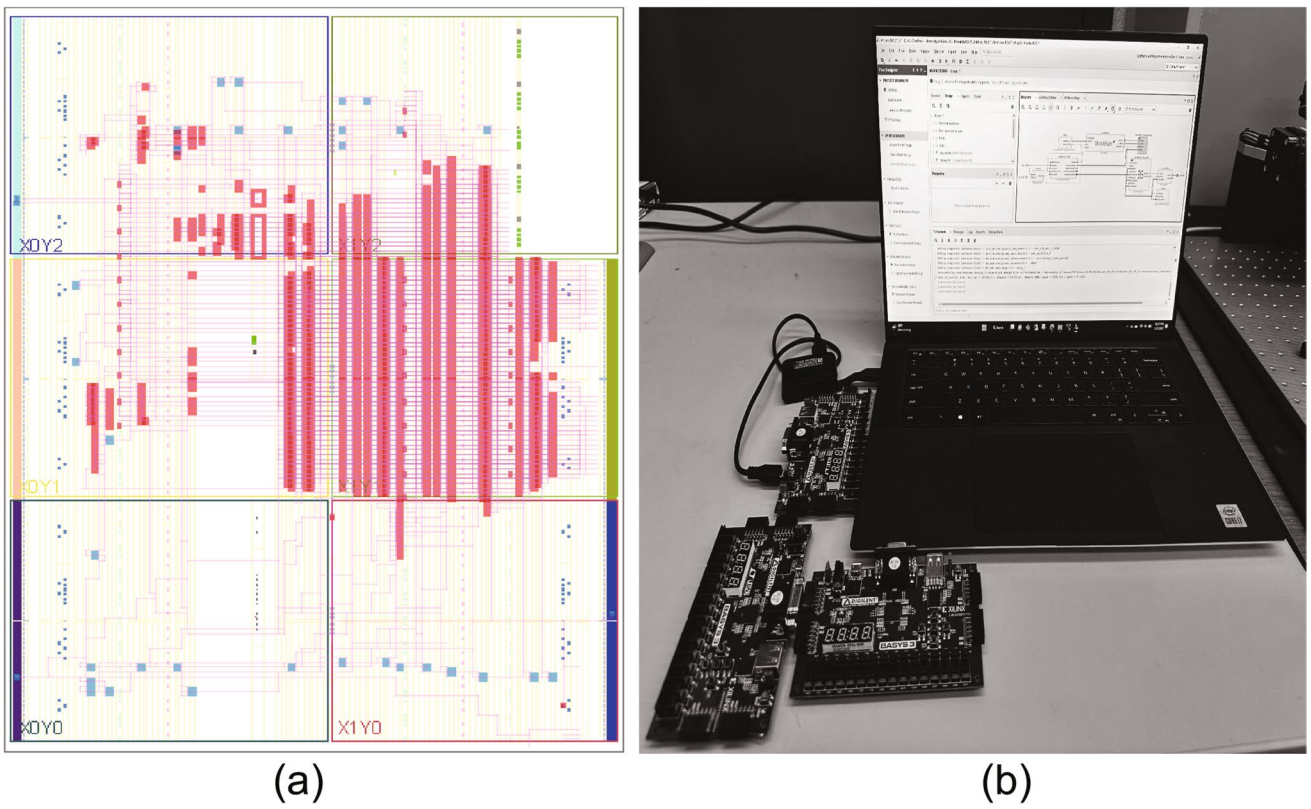


Fig. 3 Experimental hardware setup. (**a**) the layout of the hardware Trojan included chip and (**b**) FPGA and trace collection setup

a representative chip per cluster to be fully analyzed. The present framework extends this line of work by combining clustering with a localization algorithm, reducing the reverse-engineering scope to a specific on-chip region rather than a whole die.

3 Experimental Setup

3.1 RO Setup

NAND gate-based ROs are generally built of N stages of inverters in the circuits. The N must be an odd integer to generate the oscillation. In this study, NAND and NOT-based ROs with N = 5 were implemented, and their outputs were captured with an 8-bit counter referenced to the system clock. The Enable signal of ROs simultaneously enables the frequency generation of ROs and counts the number of cycles in the counter. After counting the frequency cycles of ROs, the traces have been sent to the connected devices using the UART module (shown in Fig. 4). The RO blocks are placed in different positions of FPGA die around the circuits. This placement allows detection of Trojan activity across multiple power rails and facilitates the subsequent localization step. Therefore, if adversaries insert Trojan hardware in any place of the chip, that will cause a voltage drop. The certain additional voltage drop (V_{mali_drop}) will affect the Eq. 1 and hence the modified equation will be,

$$f = \frac{\mu_g \times (V_{DD} - V_{threshold} - V_{mali_drop})^{\alpha}}{2N \times k} \quad (3)$$

Equation 3 shows that the frequency of the ROs gets affected by the insertion of hardware Trojan. The placement of the RO sensors plays an important role in the spatial resolution of the proposed localization algorithm. In this work, the ROs are not placed randomly. Instead, they are distributed across the FPGA fabric to provide spatial coverage around the protected logic region. The objective is to ensure that localized power-grid disturbances caused by Trojan activity are reflected in the frequency response of one or more nearby ROs. Since a Trojan increases switching activity and introduces additional localized voltage drop, ROs located closer to the affected region are expected to show larger frequency deviations than distant ROs.For the FPGA prototype, the ROs were placed using fixed physical constraints so that their locations remained consistent across different Trojan-free and Trojan-inserted configurations. The placement and routing of the ROs were constrained to reduce implementation-dependent variation and to make the measured frequency changes more attributable to Trojan-induced disturbances rather than synthesis or routing randomness. In the current implementation, the RO array provides coarse-grained localization, where the Trojan location is inferred relative to the nearest affected RO. Therefore, the proposed method does not claim exact gate-level localization. Instead, it identifies a smaller region of interest around the most affected RO or set of ROs, thereby reducing the amount of reverse engineering required for confirmation.

3.2 Trojan Design

As shown in Fig. 5, both combinational and sequential Trojan circuits were implemented to evaluate the framework.

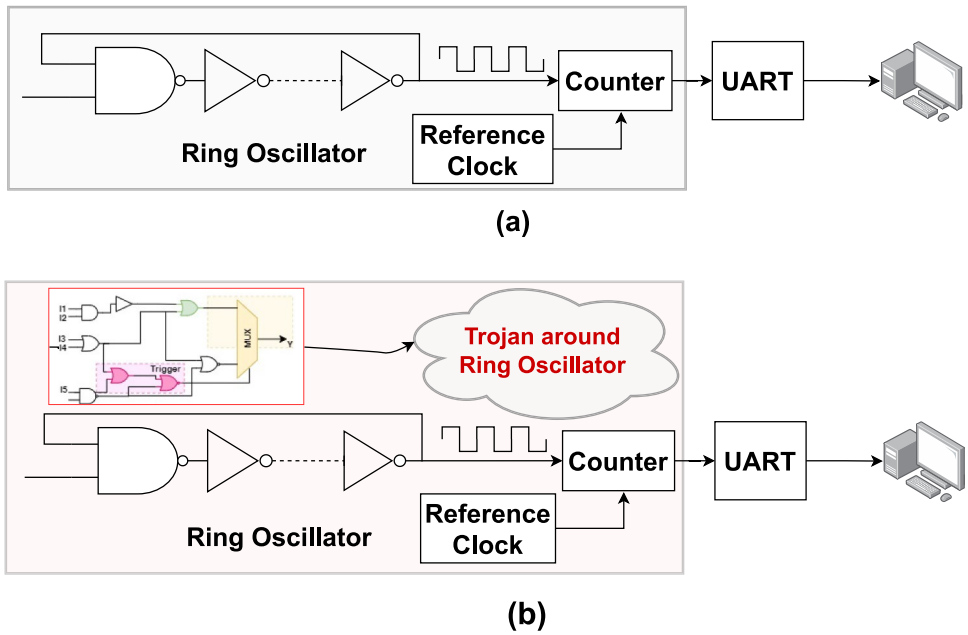


Fig. 4 Experimental circuit setup. (**a**) RO-based protection without hardware Trojan and (**b**) RO-based protection with hardware Trojan

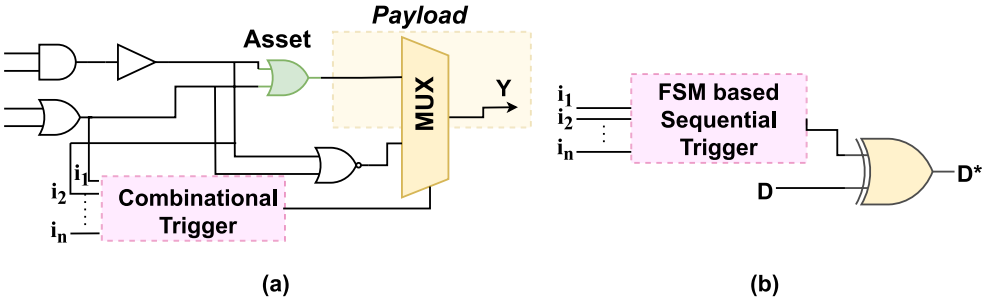


Fig. 5 Experimental Trojan circuit setup with N inputs. (**a**) Combinational Trojan and (**b**) FSM-based Sequential Trojans with K states

The combinational Trojans were designed to activate under specific logic conditions and leak cryptographic assets, while sequential Trojans were triggered through finite state machines (FSMs). For each category, payload sizes of 8, 16, and 32 bits were inserted. Custom Trojans of controlled size and activation behavior were designed to provide repeatable proof-of-concept evaluation.

These Trojans alter switching activity and increase localized power consumption, leading to measurable changes in RO frequency. Importantly, the framework does not separately detect "trigger" and "payload" circuits; instead, it captures their combined impact on the side-channel.

In the experiment, ROs are implemented with an md5 cryptographic core and a Microblaze processor. The setup is maintained with Trojan circuits in a deactivated state to collect Trojan-free RO data. Ensuring the consistent experimental setup, with identical placement and routing, the Trojans were then activated, and data impacted by the Trojan was collected.

3.3 FPGA Setup

Three Basys 3 FPGAs are used to implement the MicroBlaze microprocessor with ROs and hardware Trojans. Each board provides an independent physical instance, ensuring that process and placement variations are not limited to a single device. Basys 3 FPGA development boards have Xilinx Artix-7(XC7A35T-1CPG236C) FPGA chips with USB-JTAG protocols, which help the board communicate with other devices (Laptop in this experiment). We have utilized the Xilinx XDC macro feature to place the ROs in a fixed position and route them identically to implement RO in the FPGA die, Fig. 3(a) shows the mapping of our RO network in, and the power region of the FPGA chip with MicroBlaze processor. It should be noted that FPGA-based results do not fully replicate ASIC-level power distribution networks, and therefore the transferability of findings to foundry-scale ICs requires further study. In particular, factors such as temperature variation and process variability were not exhaustively explored in this work, and are acknowledged as limitations.

4 Analysis Methods

For the examination of Trojan variations, both combinational and sequential Trojans were implemented with N inputs and K states for sequential Trojans, as depicted in Fig. 5. The values for N and K were comprehensively explored, including $N, K \in \{8, 16, 32\}$. This systematic approach has enabled an analysis of the impact of different kinds of hardware Trojans on the RO data, significantly advancing the field of research (Figs. 6 and 7).

4.1 Overview

The detection framework leverages unsupervised ML clustering to separate Trojan-free and Trojan-inserted chips. The research aims to accurately identify trojan-infected chips by leveraging an unsupervised ML approach. For every chip, a test configuration, including trojan-free and infected samples, is used to gather data. The collected data undergoes several steps, as illustrated in Fig. 8, starting from preprocessing and transformation into a sequential format. Features are extracted from the sequential data based on the amplitude of the sequence. The clustering algorithm categorizes the chips into trojan-infected or trojan-free clusters. The AI model is optimized for precise categorization to find the ideal number of clusters.

Upon classifying the chips based on the clusters assigned by the clustering algorithm, the sequential data of the chips is passed through a proposed localization model. Our localization algorithm sorts the list of the ROs within each chip, providing the position at which the Trojan is inserted. This knowledge enables reverse engineering in the specific Area of the chip, thereby saving time. After determining if the FPGA chip is infected by trojan-infected, the clusters generated by the unsupervised model are further classified to identify which clusters contain infected chips and which are free of trojans. This comprehensive approach allows for accurately classifying chips with and without Trojan insertions.

The labeled data is utilized not for training the model but for assessing the accuracy of the gold-standard-free approach to trojan detection to evaluate the effectiveness

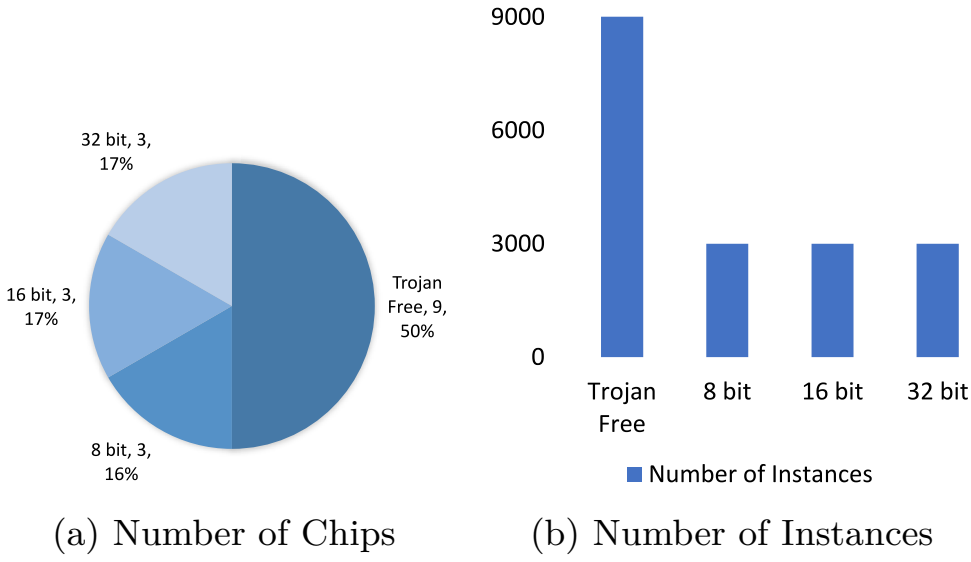


Fig. 6 Exploratory data analysis: Comparison of number of chips and instances of various Trojan sizes

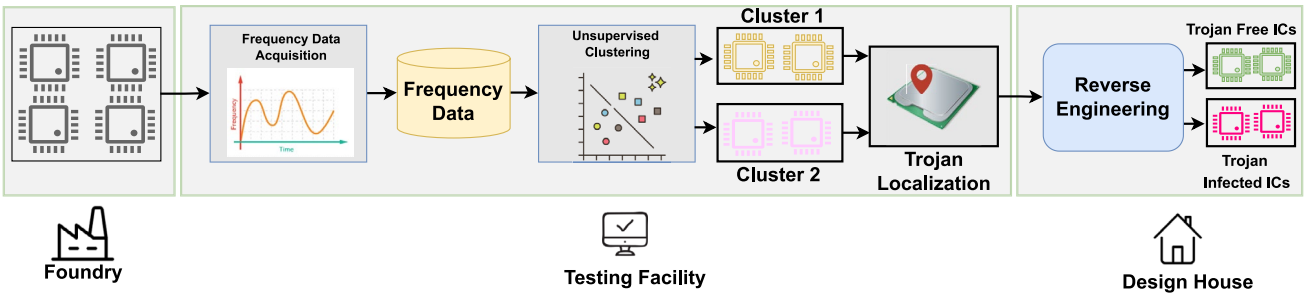


Fig. 7 Framework of the proposed method

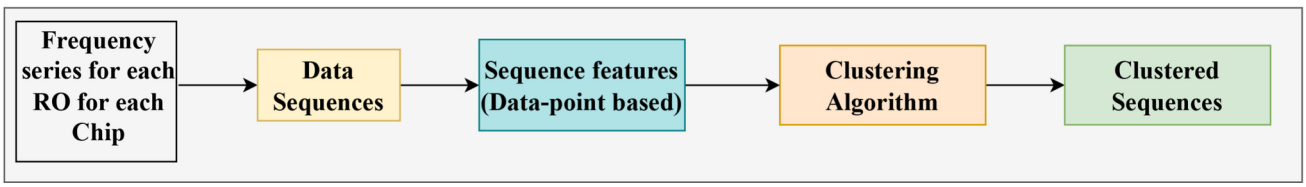


Fig. 8 RO frequency sequence clustering workflow for golden-chip-free Trojan detection. RO frequency traces are converted into data-point-based sequence features and clustered to separate Trojan-free and Trojan-inserted behavior

of our approach. Evaluation metrics include Precision, the Area Under the Curve (AUC) value, Accuracy, F1-score, False Negative Rate (FNR), False Positive Rate (FPR) and Recall, which are used to ensure the detection process's accuracy and mitigate the risk of misidentifying Trojan-free integrated circuits as infected, Fig. 7 delineates the procedural sequence of the unsupervised machine learning methodology applied for the identification of hardware Trojans within semiconductor integrated circuit (IC) supply chains.

4.2 Data Collection and Preprocessing

In this study, the reliability of the unsupervised clustering results was contingent on the meticulous data preparation process. Each integrated circuit had data available for both "golden" or Trojan-free samples and samples containing Trojans but only labeled for evaluation. Table 1 concisely summarizes the collected data.

Table 1 Overview of dataset characteristics

Chip Count	18
Count of ROs per chip	32
Input Instances per chip	1001
Chips without Trojan contamination	9
Chips Affected by Trojans	9

In Fig. 6(a), a pie chart presents the exploratory data analysis based on the number of chips with different Trojan sizes. Additionally, Fig. 6(b) displays a bar graph representing the exploratory data analysis for the number of instances corresponding to various Trojan sizes. Preprocessing techniques

eliminate redundant or extraneous features, standardize feature values, and identify and address missing data or outliers using imputation and the interquartile range.

These procedures facilitated the generation of a dataset suitable for feature extraction and clustering analysis. These specific subjects will be addressed in the section that follows:

4.2.1 Data Cleaning

Data cleaning is identifying and removing data inaccuracies, inconsistencies, and outliers. In this study, the moving average filter is used to remove random fluctuations in data points caused by noise during the experimental process in the lab. The moving average filter is a smoothing technique used to smooth out short-term fluctuations in data by taking the average of a set of data points over a certain period.

4.2.2 Normalization

The amplitude of each RO was scaled by taking the average of all chips' frequency values for each RO and dividing it by the average of all frequency values for each RO. The data is transformed to allow for better analysis and comparison of the frequency values across all chips and RO.

4.3 Sequence Formatting

The frequency series from different inputs in each RO is initially taken as the input data sequences. These sequences exhibit varying amplitudes due to the presence of trojans, which cause fluctuations in frequency as a result of power consumption. The data sequences are of a predetermined length equal to the number of input instances provided in the experiment and are kept constant for all chips. Each RO's frequency sequence was represented as a one-dimensional array of 1001 samples, forming the input to clustering. The total sequence inputs are equal to the number of chips multiplied by the number of ROs in each chip. In this case, the number of sequence inputs is 567.

Let's denote the raw frequency series for the ith Input of an RO as $F_i(t)$, where t represents time. The series can vary due to the presence of Trojans. The mathematical representation of this is given by:

$$F_i(t) = \begin{cases} f_i(t) & \text{if trojan-free} \\ f_i(t) + \delta(t) & \text{if trojan-infected} \end{cases} \tag{4}$$

Where:

- $f_i(t)$ represents the normal frequency series for the ith input.
- $\delta(t)$ represents the fluctuation in frequency due to the presence of the Trojan.

4.4 Sequence Feature Extraction

To capture the significant events in the data sequence, the amplitude of spikes extracted from $F_i(t)$ at various time instances can be denoted as $A_i(t)$. Each sequence is represented as a high-dimensional vector space as follows:

$$S_i = [A_i(t_1), A_i(t_2), \ldots, A_i(t_n)] \tag{5}$$

Where:

- $t_1, t_2, \ldots, t_n$ are the specific time instances when spikes are observed.
- n is the predetermined length of the data sequences, equal to the number of input instances provided in the experiment.

Furthermore, to determine the exact positions of these spikes within a sequence, a mapping function can be introduced:

$$P_i = [p(A_i(t_1)), p(A_i(t_2)), \ldots, p(A_i(t_n))] \tag{6}$$

Where:

- $p(A_i(t_j))$ gives the position of the jth spike in the ith sequence.

4.5 Clustering Model

After the features are extracted from the data sequences, the sequences are thus clustered using clustering techniques based on the extracted information. Five clustering techniques are implemented to compare the performance and find the best model to make the experiment fair. The goal was not to propose a new clustering technique, but to evaluate which existing model best fits golden-free Trojan detection.

4.5.1 K-means Clustering

K-means clustering is used to ascertain the cluster centroids. The Elbow method, a commonly used technique in cluster analysis, is applied to determine the optimal number of clusters. This method involves plotting the within-cluster sum of squared distances against the number of clusters and selecting the point of inflection on the curve as the ideal number. Each cluster corresponds to a particular data category, with one cluster representing trojan-free data and the other representing trojan-inserted data.

The K-means algorithm partitions the sequence features of the data sample into a certain number of clusters. Random data points are chosen as the initial cluster of centroids

or seeds. Our approach seeks to reduce the sum of squared distances between individual cluster centroids and data points through iterative updates of the centroids. This iterative process's primary objective is optimizing the clustering solution. Cluster centers (centroid) are then computed by determining the mean of all data points within that cluster. After each iteration, we compare the newly obtained cluster centroids with those from the previous iteration. If the centroids are within a defined threshold in Euclidean space, indicating their proximity, we terminate the iteration, considering the new centroids as the final centroids. Otherwise, we continue the iteration process, utilizing the new centroids as seeds for the subsequent iteration of K-means clustering.

Upon convergence of the iteration process, the data points are classified based on their association with the clusters derived from the final K-means clustering step. This classification enables the distinction between data containing trojans and trojan-free data, providing crucial insights for trojan detection in applications. The mean squared error (MSE) function serves as a statistical measure to assess the effectiveness and quality of clustering. The MSE is calculated as follows [25].

$$J = \sum_{q=1}^{K} \sum_{p=1}^{N} ||a_p^q - c_q||^2$$

Where:

- K represents clusters number.
- N represents cases number.
- $||a_p^q - c_q||^2$ is the distance function, representing the squared Euclidean distance between the pth case a_p^q and the centroid c_q for cluster q [25].

- a_p^q is the pth case in cluster q.
- c_q is the centroid for cluster q.

4.5.2 BIRCH Clustering

Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) is an alternative clustering algorithm to classify data points into trojan-free and trojan-inserted classes. BIRCH offers the advantage of scalability and efficiency when dealing with large datasets.

Figure 9b depicts the Clustering Feature (CF) tree structure for determining the optimal number of clusters in the BIRCH algorithm. The CF tree, constructed iteratively, provides insights into the distribution of data points based on predefined thresholds and aids in identifying the ideal number of clusters. Each CF tree node includes information such as the radius, centroid, and quantity of data points for its subcluster. The BIRCH algorithm is guided by the branching factor, which dictates the maximum number of subclusters are merged into a single node. This parameter gradually influences the combination of subclusters within the CF tree until the number of clusters is achieved.

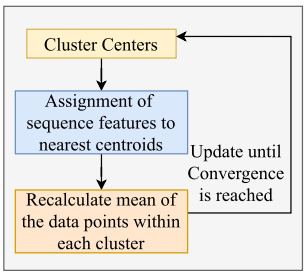
After the CF tree construction, the data points are classified based on their proximity to the subclusters. Data points closer to the centroids of trojan-free subclusters are classified as trojan-free, while those closer to the centroids of trojan-inserted subclusters are classified as trojan-inserted.

The BIRCH clustering quality is evaluated using the CF compactness measure. It is calculated as the average distance of each data point to its subcluster's centroid [26]:

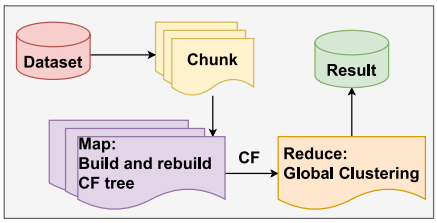
$$CF = \frac{1}{N} \sum_{p=1}^{N} \text{dist}(a_p, c_p) \tag{7}$$

Where:

- N is the total number of data points.



(a) Block diagram showing the mechanism of KMeans Clustering



(b) Block diagram showing mechanism of BIRCH

Fig. 9 Block diagrams of the two representative clustering mechanisms used in the proposed golden-chip-free Trojan detection framework: (**a**) K-means clustering and (**b**) BIRCH clustering. K-means iteratively assigns RO frequency sequence features to the nearest cluster centroids and updates the centroids until convergence. BIRCH incrementally builds and refines a clustering feature (CF) tree from data chunks and then performs global clustering on the reduced CF representation

- $\text{dist}(a_p, c_p)$ represents the Euclidean distance between the pth data point a_p and its subcluster's centroid c_p, given by [26]:

$$\text{dist}(a_p, c_p) = \sqrt{\sum_{q=1}^{D}(a_p^q - c_p^q)^2}$$

- D is the number of features or dimensionality of the data.
- a_p^q is the qth feature of the pth data point.
- c_p^q is the qth feature of the centroid for the subcluster of the pth data point.

4.5.3 Agglomerative Clustering (AGNES)

AGNES is an alternative approach used to classify data points into trojan-free and trojan-inserted classes. At the outset, each point is considered a separate cluster, and through successive iterations, the more akin clusters are merged till the desired number of clusters is achieved.

Different linkage criteria, such as complete linkage, average linkage, and Ward's approach, are employed to measure the similarity between clusters [27]. Total linkage calculates the most significant distance between data points in different clusters. Average linkage involves computing the average distance between each pair of data points from distinct clusters. In Ward's approach, cluster merging aims to minimize the overall increase in within-cluster variance.

The AGNES procedure is iteratively applied until the number of clusters is reached. At each stage, the two closest clusters are combined, creating a hierarchical structure called a dendrogram. This dendrogram, reflective of the required level of similarity, facilitates the determination of the optimal number of clusters and provides insights into the clustering process. Following the agglomerative clustering process, data points are classified based on their membership in the trojan-free or trojan-inserted clusters.

4.5.4 Gaussian Mixture Model (GMM) Clustering

GMM clustering is another viable technique for trojan detection, distinguishing between trojan-free and trojan-inserted data points. GMM represents the probability distribution of the data using a mixture of Gaussian distributions.

The GMM clustering algorithm employs an expectation-maximization (EM) technique, initially estimating parameters of Gaussian distributions, including mean vectors and covariance matrices. Iteratively optimizing the likelihood of the data, the EM algorithm refines its estimations of the parameters based on the current values.

Once the parameters are estimated, the data points are assigned to the Gaussian components that maximize their likelihood. Each Gaussian component represents a trojan-free or trojan-inserted cluster. By assigning the data points to the most likely Gaussian component, we classify them accordingly.

The likelihood of a data point a_p belonging to a Gaussian component k is computed using the formula:

$$P(x_i|k) = \frac{1}{(2\pi)^{D/2}|\Sigma_k|^{1/2}} \times \exp\left(-\frac{1}{2}(x_i - \mu_k)^T \Sigma_k^{-1}(x_i - \mu_k)\right) \quad (8)$$

Where:

- D is the dimensionality of the data.
- μ_k is the mean vector of the kth Gaussian component.
- Σ_k is the covariance matrix of the kth Gaussian component.

4.5.5 Hidden Markov Model (HMM) Clustering

HMM clustering provides an alternative approach for trojan detection by capturing temporal dependencies within sequential data. HMMs consist of a set of hidden states representing different underlying conditions or clusters and observable symbols representing the observed data points.

The HMM clustering algorithm involves two main processes: the training and classification phases. In the training phase, the parameters of the HMM, including the transition probabilities between states and the emission probabilities of observable symbols, are estimated using the Baum-Welch algorithm.

4.6 Golden Free Approach

The detection approach used in this work follows a golden-chip-free methodology, where the clustering model does not require a known Trojan-free chip, a golden side-channel signature, or labeled training data during detection[17]. Unlike traditional side-channel detection methods that compare measurements from suspect chips against trusted golden ICs, the proposed framework measures RO frequency data from a population of suspect chips and analyzes their similarity through unsupervised clustering. Therefore, no chip is assumed to be trusted before clustering, and the clustering model is trained only on the measured RO frequency behavior of the available chip population. [10].

While Let D = $\{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_q\}$ denote the set of RO frequency measurements collected from q chips, where Xi

represents the preprocessed RO sequence features of the ith chip. The clustering model partitions D into C clusters.

$\mathcal{D} \rightarrow \{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_C\}$, where chips assigned to the same cluster are expected to have similar RO side-channel characteristics. If Trojan-induced switching activity or localized voltage drop significantly changes the RO frequency response, Trojan-inserted chips are expected to form clusters that are separable from Trojan-free chips. After the clusters are formed, physical interpretation of each cluster may require confirmation. Similar to prior golden-free clustering-based approaches, one representative chip from each cluster can be selected for destructive reverse engineering or detailed physical inspection. This step is not used to train the clustering model and does not serve as a golden reference for the remaining chips. Instead, it is a post-clustering validation step used to assign a semantic label, such as Trojan-free or Trojan-inserted, to an already formed cluster. Once the representative chip from a cluster is analyzed, other chips in the same cluster are inferred to share the same security status because their RO side-channel responses are similar.This distinction is important. The proposed method reduces the dependence on destructive reverse engineering, but it does not claim to eliminate physical confirmation entirely. In conventional golden-chip-based approaches, destructive analysis may be required to establish a trusted reference or to compare many suspect chips against a known-good signature. In contrast, the proposed framework requires confirmation only at the cluster level. If C clusters are generated, at most one representative chip per cluster needs to be physically inspected. Thus, the reverse-engineering effort is reduced from inspecting a large population of chips to inspecting a small number of representative cluster samples.In addition, the proposed localization algorithm further reduces the physical verification effort. Instead of performing full-chip reverse engineering on the representative chip, the RO localization score ranks the RO neighborhoods according to their Trojan-impact levels. The highest-scoring RO regions identify the most likely physical area affected by Trojan-induced power-grid disturbance. Therefore, physical inspection can focus on a smaller region of interest around the most affected ROs rather than the entire die.In the experimental evaluation, labels indicating Trojan-free and Trojan-inserted configurations are used only to compute performance metrics such as accuracy, precision, recall, false positive rate, and F1 score. These labels are not used during clustering or localization. Therefore, the reported results evaluate how well the golden-free clustering process separates the chip population without relying on labeled training data or a pre-characterized golden chip.

4.7 Proposed Algorithm for Localization of Trojan

In this research paper, we propose a novel approach for localizing Trojans within ICs based on the analysis of RO data. By accurately identifying the potential location of the Trojan, our approach aims to reduce the verification time and cost associated with reverse engineering while enhancing the detection confidence. This paper presents a comprehensive methodology for narrowing down the region of interest in suspect ICs, facilitating efficient and effective Trojan localization.

Algorithm 1 Localization of Trojan Algorithm

Require: $chipData$ - Frequencies data matrix ($m \times n$), where m represents input instances and n represents different frequencies.
Ensure: $chipDataScore$ - Sorted frequencies data matrix based on the impact of trojans.
1: $avgRowArr \leftarrow$ Average($chipData$, rows)
2: $avgColArr \leftarrow$ Average($chipData$, columns)
3: $chipData \leftarrow$ Normalize($chipData$, $avgRowArr$, $avgColArr$)
4: $chipDataArr \leftarrow$ SplitIntoChunks($chipData$, 10)
5: $medianArr \leftarrow$ ComputeMedians($chipDataArr$)
6: $avgMed \leftarrow$ Average($medianArr$)
7: $devArr \leftarrow$ ComputeDeviations($chipDataArr$, $avgMed$)
8: $chipDataScore \leftarrow$ Sort($chipData$, $devArr$)

In the given algorithm 1, *chipData* represents the input frequencies data matrix with dimensions $m \times n$, where *m* represents the number of input instances, and *n* represent different frequencies. The algorithm aims to localize the trojan impacts within the chip data. It first computes the average row and column values in the input matrix normalizes the data and splits it into chunks. It calculates median values and deviations for each chunk and then sorts the rows based on these deviation values. Finally, the algorithm computes the sum of deviation values for each row, sorting the rows in *chipData* based on these sums to obtain *chipDataScore*.

This algorithm systematically identifies and localizes the impact of trojans in the given chip data by focusing on distinct patterns within frequency sequences. It achieves this through mathematical operations, medians, and deviations, ensuring an effective sorting mechanism to prioritize rows with significant trojan impacts in the final *chipDataScore* output.

The localization accuracy depends on the density, placement, and spatial coverage of the RO network. A denser RO array can provide finer localization granularity because each Trojan-induced disturbance is more likely to be captured by a nearby sensor. Conversely, if the ROs are sparsely placed or if a Trojan is located far from all ROs, the localization result becomes coarser and may only indicate the broad affected region. For this reason, the output of the localization algorithm should be interpreted as a ranked list of RO

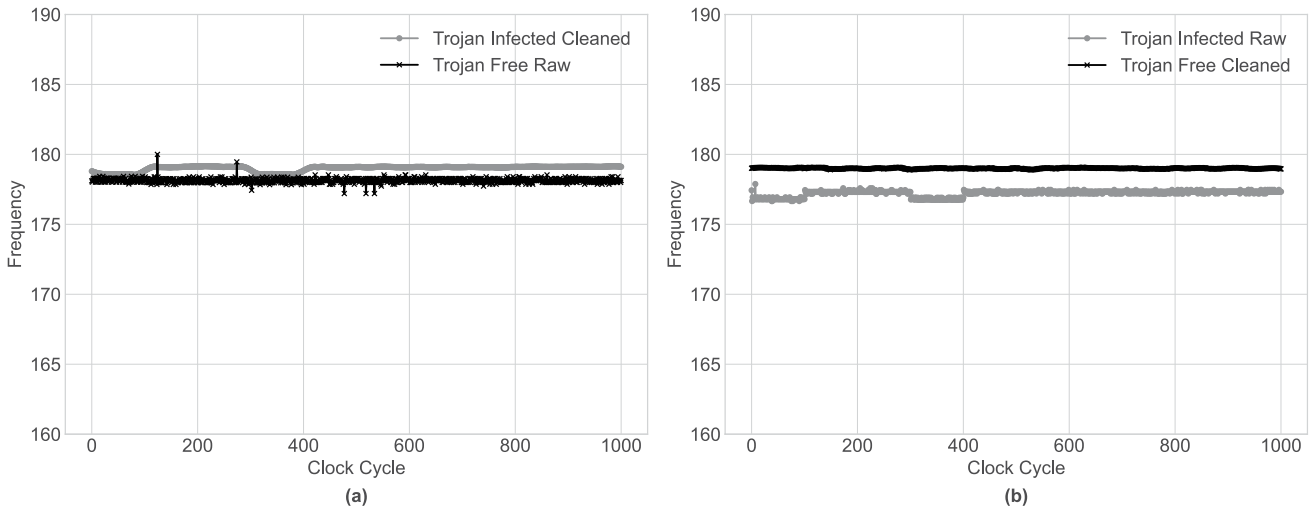


Fig. 10 Analyzing Frequency Series Measurements (Sample RO1) Before and After Data Cleaning in Trojan-Infected and Trojan-Free Chips

neighborhoods rather than an exact physical coordinate of the Trojan. In the FPGA experiments, the Trojan was intentionally inserted near RO1, and the highest localization score was consistently assigned to RO1, demonstrating that the proposed scoring method can identify the nearest affected RO under the evaluated placement strategy.

5 Results

5.1 RO Data Sequences and Data processing

The frequency side-channel information from various trojan configurations was measured using an RO. The data series exhibited predominantly varying frequencies due to an additional malicious circuit triggered by input signals applied to the chip. A total of 32032 measurements were conducted for each test configuration for the analysis of 32 data sequences using ML methods. Baseline measurements were also obtained for the chip without switching logic to capture unaffected frequencies. These measurements were a reference for training the ML model to detect deviations caused by trojan-infected configurations.

The frequency series measurements for chips, both with and without trojans, are depicted in Fig. 10. The frequency measurements of trojan-free chips exhibit distinct similarities. In contrast, those of chips with introduced trojans reveal variations in the sequence pattern due to additional current in the power grid. The associated visuals effectively portray these distinctions in frequency measurements between trojan-free and trojan-infected semiconductors. Notably, and as anticipated, the magnitude of the difference becomes more pronounced with larger counter trojans.

A moving average filter was employed to enhance the quality of the data sequences. Random spikes in the data could confuse the ML model when discerning frequency patterns. However, as demonstrated in Fig. 10 (b) and (d), these spikes have been effectively eliminated through the application of the moving average filter for both Trojan-inserted and Trojan-free chips. Consequently, the resulting cleaner samples facilitate more accurate identification of deviations by the ML model.

5.2 Usage of Clustering

In the practical detection scenario, a crucial aspect is utilizing clustering techniques to analyze the collected data. The resulting cluster maps for each data set provide valuable insights into the potential number of chips present, enabling a more meticulous analysis of representative chips from each cluster. Figure 11 demonstrates a scatter plot of clustering involving multiple chip configurations. The algorithm, detailed in Section IV, effectively separates the data into distinct clusters. The application of Principal Component Analysis (PCA) facilitates the clustering process. This method

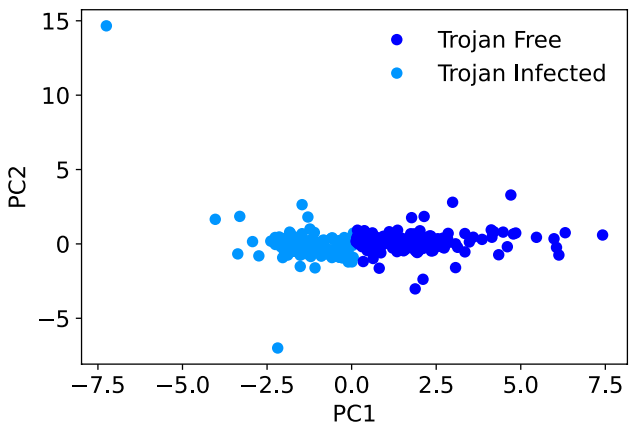


Fig. 11 Scatter plot showing the clusters of Trojan infected and Trojan free data with PCA

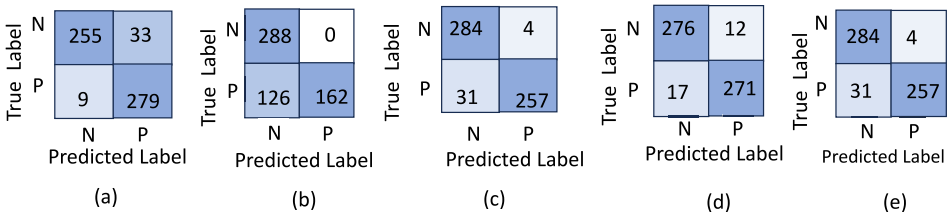


Fig. 12 Confusion matrix illustrating the classification results of various Clustering Models for overall detection of trojan-infected and trojan-free instances (**a**) AGNES, (**b**) BIRCH, (**c**) GMM, (**d**) HMM, and (**e**) Kmeans Clustering

transforms high-dimensional data into a new coordinate system, effectively reducing its dimensionality. The primary components establish the new coordinate system, aligning with the directions of the highest variation in the data. Our approach uses PCA to plot the image clusters, utilizing two principal components. By reducing the dimensionality of the image data to two dimensions, the data points are plotted according to their positions in the new coordinate system.

The resulting plot effectively displays the distribution of the images across the clusters, providing a visual representation of the similarity between images within the same cluster. This visualization aids in understanding the patterns and relationships among the images, facilitating the identification of shared characteristics and potential trojan-infected chips. Figure 11 is an exemplar, showcasing the image clusters and offering valuable insights into the detected patterns. Using clustering techniques and PCA, the analysis of chip configurations becomes more efficient and allows for a comprehensive understanding of the data.

5.3 Detection of Trojan

5.3.1 Results and Evaluation of Hardware Trojan Detection for Trojan-Free and Trojan-Inserted Chips

The proposed unsupervised ML-assisted SCA-based Trojan detection technique achieved remarkable accuracy in identifying Trojans of various sizes.

An assessment is conducted on an FPGA-based implementation of an md5 cryptic core equipped with an RO network intentionally embedded with Trojans of varying sizes. The results indicate that the proposed method effectively detects Trojans without relying on golden data. Examining visualized clusters helps understand their structure while analyzing the confusion matrix, which provides a detailed breakdown of how well the algorithm classifies data.

Figure 12 displays the confusion matrix, providing a comprehensive overview of multiple models to cluster frequency sequences for distinct inputs measured by each RO in every chip. It illustrates accurate and inaccurate classifications for both Trojan-free and Trojan-infected instances. In the figure, 'N' represents Trojan-free, and 'P' represents Trojan-infected. Matrix entries enable assessing the method's resilience and correctness in cluster assignments, making the confusion matrix a valuable tool for evaluating the algorithm's ability to distinguish between groups.

A comparative evaluation of all clustering models is summarized in Table 2 and Fig. 13. The results show that HMM achieved the highest overall accuracy (0.95) and F1 score (0.949), reflecting its strength in modeling sequential RO data. K-means and GMM also performed competitively, with accuracies above 0.93 and very low false positive rates (0.013). AGNES achieved strong recall (0.96) but at the cost of a higher false positive rate (0.11). BIRCH, while scalable, performed the weakest with accuracy of 0.781 and recall of only 0.56, highlighting its limitations for fine-grained Trojan detection. Precision remained consistently high across most models, especially K-means (0.987) and GMM (0.984), indicating that Trojan detections are rarely false alarms. The comparative study demonstrates that multiple unsupervised clustering techniques are viable for golden-free detection, with HMM providing the best trade-off between accuracy, recall, and localization capability.

5.3.2 Results and Evaluation of Hardware Trojan Detection Based on Trojan Size

The results of the comparison of Accuracy, Recall, and Macro F1 score for different sizes of Trojans using sequence clustering with the HMM model are presented in Fig. 14a. Considering the different sizes of Trojans, the confusion matrix for multi-class classification is provided in Fig. 14b.

The detection algorithm's accuracy is evaluated for various Trojans, ranging from Trojan-free chips to Trojans with 8, 16, and 32 bits. The Trojan-free chips demonstrated outstanding accuracy, achieving 98.6%. This high accuracy confirms the reliability of the proposed method in distinguishing between chips without Trojans and those containing Trojans.

Table 2 Table showing Performance Metrics for overall Trojan detection using Clustering techniques

Metrics	Kmeans	AGNES	BIRCH	GMM	HMM
Accuracy	0.941	0.927	0.781	0.939	0.95
F1 Score	0.938	0.93	0.72	0.936	0.949
Precision	0.987	0.894	1.0	0.984	0.957
Recall	0.895	0.96	0.56	0.892	0.94
FPR	0.013	0.11	0.0	0.013	0.041
FNR	0.104	0.03	0.437	0.107	0.059

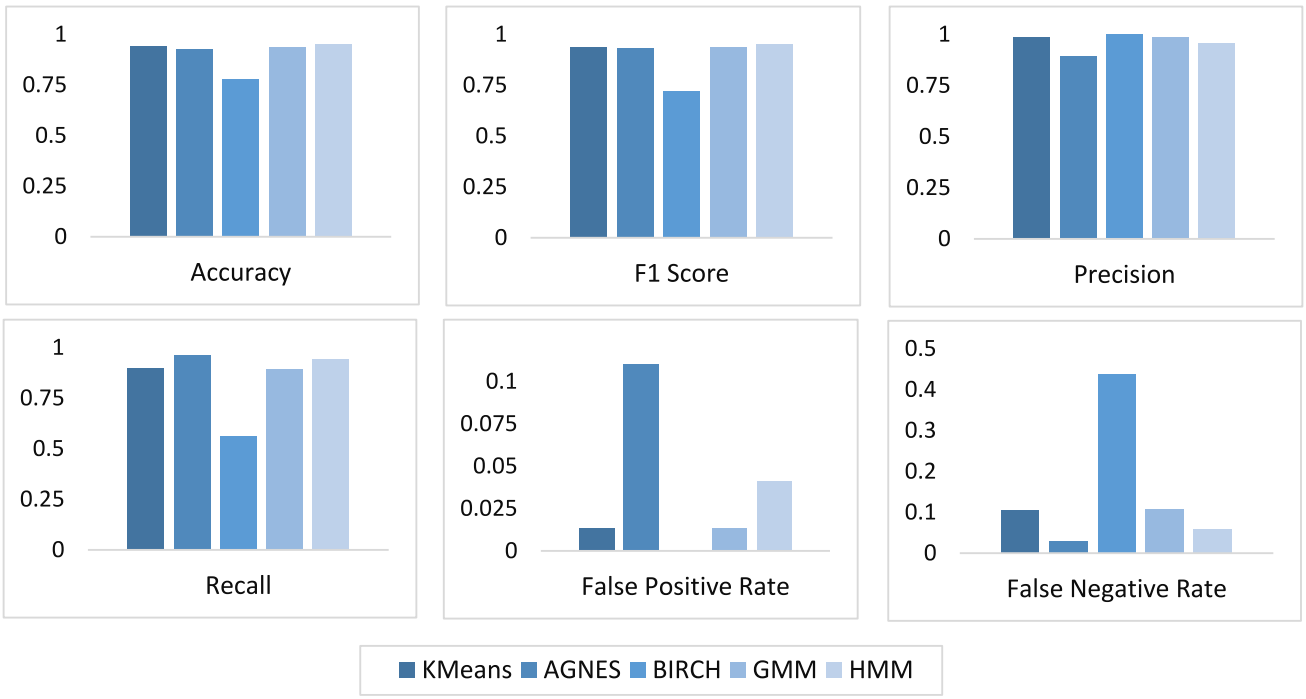
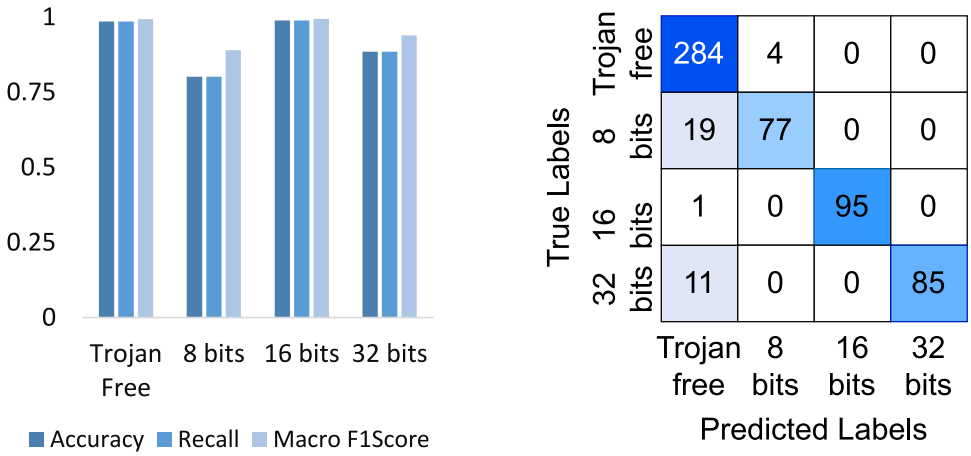


Fig. 13 Performance metrics comparison for various clustering techniques

Fig. 14 Comparison of HMM model performance with different sizes of Trojan using sequence clustering



(a) Performance Metrics (b) Confusion Matrix

For Trojans with 8 bits, the detection accuracy is measured at 80.2%. This value is slightly lower than the Trojan-free case as the number of Trojan-free samples was higher. This result highlights the ability of the proposed technique to detect small Trojans effectively. The accuracy increased to 98.9% for Trojans with 16 bits, further validating the efficacy of the proposed method in detecting more prominent Trojans. The detection accuracy remained consistently high, even for Trojans with 32 bits, reaching 88.5%. This demonstrates the robustness of the method in identifying more prominent Trojans. Also, additional metrics such as recall and F1 score are measured to provide a comprehensive evaluation. The recall values are calculated and aligned with the accuracy results, indicating that the proposed method successfully identified Trojans across different sizes. The F1 scores, which consider both precision and recall, consistently exhibited high values above 0.89 for all Trojans, further affirming the reliability of the proposed approach.

In summary, the experiments show that the clustering-assisted framework achieves high detection accuracy of hardware Trojans. Across five clustering models (K-means, AGNES, BIRCH, GMM, and HMM), HMM achieved the best detection rate of 95%. Other models also provided reliable separation between Trojan-free and Trojan-inserted

chips. The contribution is not the proposal of HMM as a novel method, but rather the demonstration that multiple clustering approaches can be adapted for golden-free detection.

5.4 Localization of Trojan in FPGA

After passing the data sequences through the sequence clustering model, the inputs are divided into 2 clusters. Each cluster's data sequences were then processed using the proposed localization algorithm. Figure 15b shows the scatter plot of the corresponding values provided by the algorithm for each data sequence of various chips, according to their RO number, for cluster 1. Similarly, Fig. 15a shows a similar plot for the data sequences in the other cluster.

In Fig. 15b, it can be observed that the plotted data points are not significantly deviated from the remaining data points, except for three outliers (note that outliers are caused by noise during frequency readings in the experiment). On the other hand, Fig. 15a reveals that the data points in RO1 and those closer to RO1 are highly deviated from the data points in other sequences of the corresponding chips. This observation suggests that the Trojan is localized near the RO1 position of the chips grouped in Cluster 2, and Cluster 1 can be labeled as trojan-free.

The detailed localization results obtained from our algorithm for the inserted Trojan near the ROs of each chip are presented in Table 3. The specific positions of the ROs in the Table are denoted by numbers 1, 2, and so forth. During the experiment, the Trojan is intentionally placed near RO 1. Upon analyzing the Table, it is evident that for each chip, the Trojan was located near RO1 as intended. However, there were some discrepancies in accurately identifying the second and third positions. Despite these errors, most of the answers provided by the algorithm remain correct, demonstrating the overall accuracy of our proposed localization approach.

The ROs with a more minor impact from the Trojan could be RO2, RO0, and RO3. Although there were errors in identifying these positions, the overall localization results indicate the algorithm's effectiveness. Moreover, considering the top two ROs with the highest trojan impact, the algorithm achieves an accuracy rate 0.88.

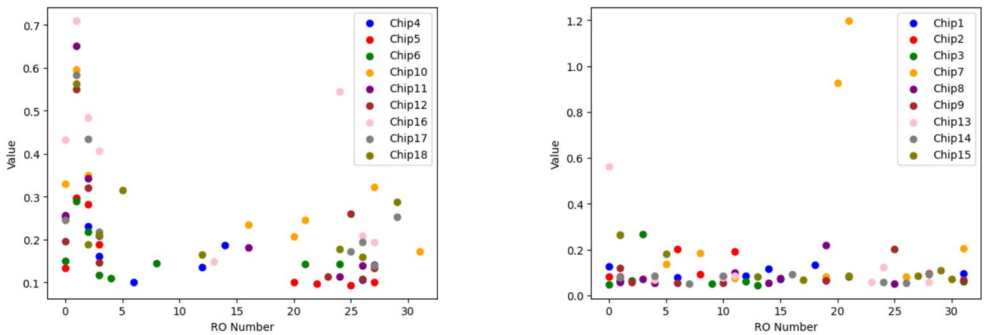
In summary, our algorithm successfully determines the precise location of the inserted Trojan by referencing the nearest RO. Most of the obtained results align with the intended placement near RO1, highlighting the accuracy of our approach.

6 Discussion

The results showcased in the preceding section underscore the effectiveness of the proposed unsupervised ML-assisted SCA-based Trojan detection technique. This section delves into a discussion and analysis of the study's key findings, implications, and limitations.

6.1 Effectiveness of the Proposed Method

The proposed method showed remarkable accuracy in identifying hardware Trojans of various sizes. The evaluation is



(a) Scatter plot showing the RO Position vs. Proposed Localization Algorithm Value for Cluster 2.

(b) Scatter plot showing the RO Position vs. Proposed Localization Algorithm Value for Cluster 1.

Fig. 15 Comparison of Trojan localization: (**a**) infected cluster vs. (**b**) Trojan-free cluster

Table 3 Localization of Trojan near ROs of each chip

	Chip4	Chip5	Chip6	Chip10	Chip11	Chip12	Chip16	Chip17	Chip18
RO	1	1	1	1	1	1	1	1	1
RO	0	2	2	2	2	2	24	2	5
RO	2	3	0	0	0	25	2	29	29

performed on an FPGA implementation of an md5 cryptic core with an RO network intentionally inserted with Trojans of different sizes. The results revealed high accuracy levels, with an overall accuracy of 94.1% and an F1 score of 93.8%

The proposed method utilizes the frequency side-channel information obtained from RO measurements. The presence of Trojans in the chip is identified by analyzing the frequency series. The use of ML algorithms, specifically K-means clustering in this study, enables the detection of deviations caused by Trojan-infected configurations. The moving average filter applied to the data sequences further enhances the accuracy by eliminating random spikes that could confuse the ML model. The results demonstrate that the proposed technique effectively captures the differences between Trojan-free and Trojan-infected chips, even for more prominent Trojans.

Furthermore, using clustering techniques, particularly PCA, provides valuable insights into the potential number of chips present and enables a more meticulous analysis of representative chips from each cluster. The scatter plot and cluster maps generated through clustering and PCA visualization facilitate the identification of shared characteristics and potential Trojan-infected chips. The visualization aids in understanding the patterns and relationships among the chips, enhancing the overall efficiency of the analysis.

6.2 Localization of Trojan

The proposed method also includes a localization component, which aims to determine the precise location of the inserted Trojan near the ROs of each chip. The results of the localization analysis indicate that the algorithm successfully identified the intended position of the Trojan near RO1 for the majority of the chips. However, some discrepancies exist in accurately identifying the second and third positions. Despite these errors, the overall localization results demonstrate the effectiveness of the proposed approach. It is important to note that the algorithm's accuracy in precisely identifying the positions near RO2, RO0, and RO3 may be lower. However, considering the top two ROs with the highest Trojan impact, the algorithm achieved an accuracy rate 88.

The results also show the importance of RO placement in the proposed localization framework. Since the algorithm ranks ROs according to their frequency deviation, the physical distance between the Trojan and the nearest RO directly affects localization accuracy. If the Trojan is placed near an RO, the corresponding sensor is expected to experience a stronger frequency shift because of local voltage drop and switching-induced disturbance. If the Trojan is located between multiple ROs, several nearby sensors may show elevated scores, producing a region-level localization result. Thus, the proposed method is best understood as RO-neighborhood localization rather than exact layout-coordinate recovery. This behavior is still useful in practice because it narrows the reverse-engineering region to a small subset of the chip instead of requiring full-chip inspection.

6.3 Limitations

Although the proposed framework demonstrates promising results, several limitations remain. First, the evaluation was conducted on FPGA boards rather than fabricated ASICs. FPGA fabrics and ASICs differ in power-grid structure, placement flexibility, routing resources, and process variation. Therefore, ASIC-level validation is necessary to quantify the transferability of the proposed RO-based localization framework.

Second, the practical insertion of ROs in ASIC designs must be considered carefully. In an ASIC setting, the RO network can be integrated by the trusted design house as part of a design-for-test, process-monitoring, or security-monitoring infrastructure. ROs and similar delay sensors are already commonly used for process, voltage, and temperature monitoring, making their inclusion practical from a design perspective. However, because the foundry is untrusted in the assumed threat model, the RO network should not be treated as a hidden or secret structure. Instead, its protection should rely on tamper-evident design principles. For example, the design house can verify the expected number of sensors, readout paths, enable behavior, and frequency response range during post-silicon test. If the foundry removes, disables, or modifies the RO network, the collected measurements would deviate from the expected sensor behavior and expose possible tampering.Third, the localization resolution depends on RO density and placement. A sparse RO network may only provide coarse region-level localization, while a denser network can improve spatial resolution at the cost of area, power, and routing overhead. Therefore, future ASIC implementations should study the trade-off among RO density, sensor overhead, power-grid coverage, localization accuracy, and resistance to foundry-side tampering. Benchmark Trojans, environmental variation such as temperature and voltage changes, and larger chip populations should also be evaluated in future work.

7 Conclusion

This paper presented a golden-chip-free hardware Trojan detection framework that integrates unsupervised clustering with distributed on-chip ring oscillator (RO) measurements. The proposed approach does not require a known Trojan-free chip, golden side-channel signature, or labeled

training data during clustering. Instead, it analyzes the similarity of RO frequency responses across a population of suspect chips and separates Trojan-free and Trojan-inserted samples based on their side-channel behavior. In addition to detection, this work introduced an RO-based localization method that ranks sensor neighborhoods according to their Trojan-impact scores. The localization step provides a coarse-grained estimate of the physical region most affected by Trojan-induced switching activity and localized voltage-drop effects. Therefore, instead of requiring full-chip reverse engineering, the proposed framework can guide physical inspection toward the most suspicious RO neighborhoods. Experimental results on FPGA prototypes demonstrate that the proposed clustering-assisted framework can achieve up to 95% detection accuracy across multiple Trojan configurations. The localization results further show that the highest-impact RO regions correspond to the intentionally inserted Trojan neighborhood, demonstrating the feasibility of combining golden-free clustering with spatial localization. The proposed method reduces the dependence on destructive reverse engineering by limiting physical confirmation to representative cluster samples and, within those samples, to smaller RO-neighborhood regions. However, the framework does not claim to eliminate physical inspection entirely. Future work will focus on ASIC-level validation, optimized RO placement strategies, robustness under process-voltage-temperature variations, benchmark Trojan evaluation, and tamper-evident RO readout mechanisms for untrusted-foundry settings.

Statements and Declarations

Funding This research was funded by a grant provided by the Air Force Research Lab (AFRL) through the Assured and Trusted Digital Microelectronics Ecosystem (ADMETE) grant, BAA-FA8650-18-S-1201, which was awarded to Wright State University, Dayton, Ohio, USA. This project was carried out under CAGE Number: 4B991 and DUNS number: 047814256.

Competing interests The authors declare no competing interests.

Author Contributions A.G. conceptualized and designed the research, managed the experimental setup, led machine learning and localization technique development, implemented algorithms, managed data preprocessing, drafted manuscript and conducted comparative analysis. T.Hossain made contributions to FPGA-based experiments and drafted the manuscript. F.A. (the corresponding author) and T.Hoque provided project supervision, offering guidance on research design and methodology. All authors collectively reviewed, revised, and consented to the final version of the manuscript for publication.

Data Availability No datasets were generated or analysed during the current study.

Ethical Approval This research did not involve the use of human or animal subjects, sensitive data, or any other ethical considerations that would require approval from an ethics committee or institutional review board.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Kumar R, Divyanshu D, Khan D, Amara S (2022) Massoud Y Spin orbit torque-assisted magnetic tunnel junction-based hardware trojan. Electronics 11(11):1753
2. Yang T, Mittal A, Fei Y, Shrivastava A (2020) Large delay analog trojans: A silent fabrication-time attack exploiting analog modalities. IEEE Transactions on Very Large Scale Integration (VLSI) Systems 29(1):124–135
3. Jain A, Zhou Z, Guin U (2021) Survey of recent developments for hardware trojan detection. In: 2021 IEEE International Symposium on Circuits and Systems (ISCAS), pp 1–5. IEEE
4. Du D, Narasimhan S, Chakraborty RS, Bhunia S (2010) Self-referencing: A scalable side-channel approach for hardware trojan detection. In: International workshop on cryptographic hardware and embedded systems, pp 173–187. Springer
5. Zhang X, Tehranipoor M (2011) Ron: An on-chip ring oscillator network for hardware trojan detection. In: 2011 Design, Automation & Test in Europe, pp 1–6. IEEE
6. Liu Y, Huang K, Makris Y (2014) Hardware trojan detection through golden chip-free statistical side-channel fingerprinting. In: Proceedings of the 51st annual design automation conference, pp 1–6. ACM
7. Hoque T, Narasimhan S, Wang X, Mal-Sarkar S (2017) Bhunia S Golden-free hardware trojan detection with high sensitivity under process noise. J Electron Test 33(1):107–124
8. Yang S, Hoque T, Chakraborty P, Bhunia S (2022) Golden-free hardware trojan detection using self-referencing. IEEE Transactions on Very Large Scale Integration (VLSI) Systems 30(3):325–338
9. Xue M, Wang J, Hux A (2016) An enhanced classification-based golden chips-free hardware trojan detection technique. In: Hardware-Oriented Security and Trust (AsianHOST), IEEE Asian, pp 1–6. IEEE
10. Faezi S, Yasaei R, Barua A, Faruque MAA (2021) Brain-inspired golden chip free hardware trojan detection. IEEE Trans Inf Forensics Secur 16:2697–2708. https://doi.org/10.1109/TIFS.2021.3062989
11. Yang S, Chakraborty P, SLPSK P, Bhunia S (2021) Trusted electronic systems with untrusted cots. In: 2021 22nd International Symposium on Quality Electronic Design (ISQED), pp 198–203. IEEE
12. Ashok M, Turner MJ, Walsworth RL, Levine EV, Chandrakasan AP (2022) Hardware trojan detection using unsupervised deep learning on quantum diamond microscope magnetic field images. ACM Journal on Emerging Technologies in Computing Systems (JETC)
13. Becker GT, Regazzoni F, Paar C, Burleson WP (2014) Stealthy dopant-level hardware trojans: extended version. J Cryptogr Eng 4(1):19–31

14. Hamlet JR, Mayo JR, Kammler VG (2019) Targeted modification of hardware trojans. J Hardw Syst Sec 3(2):189–197
15. Dai S, Rosenstein JK (2015) A 14.4 nw 122khz dual-phase current-mode relaxation oscillator for near-zero-power sensors. In: 2015 IEEE Custom Integrated Circuits Conference (CICC), pp 1–4. IEEE
16. Gravellier J, Dutertre J-M, Teglia Y, Loubet-Moundi P (2019) High-speed ring oscillator based sensors for remote side-channel attacks on fpgas. In: 2019 International Conference on ReConFigurable Computing and FPGAs (ReConFig), pp 1–8. IEEE
17. Worley K, Rahman MT (2019) Supervised machine learning techniques for trojan detection with ring oscillator network. In: 2019 SoutheastCon, pp 1–7. IEEE
18. Cui X, Wu K, Karri R (2018) Hardware trojan detection using path delay order encoding with process variation tolerance. In: 2018 IEEE 23rd European Test Symposium (ETS), pp 1–2. IEEE
19. Tehranipoor M (2016) Koushanfar F A survey of hardware trojan taxonomy and detection. IEEE Design & Test of Computers 01:1–1
20. Kelly S, Zhang X, Tehranipoor M (2015) Ferraiuolo A Detecting hardware trojans using on-chip sensors in an asic design. J Electron Test 31(1):11–26
21. Bhunia S, Hsiao MS, Banga M (2014) Narasimhan S Hardware trojan attacks: threat analysis and countermeasures. Proc IEEE 102(8):1229–1247
22. Gohil V, Guo H, Patnaik S, Rajendran J (2022) Attrition: Attacking static hardware trojan detection techniques using reinforcement learning. In: Proceedings of the 2022 ACM SIGSAC conference on computer and communications security, pp 1275–1289
23. He J, Ma H, Liu Y (2020) Zhao Y Golden chip-free trojan detection leveraging trojan trigger's side-channel fingerprinting. ACM Transactions on Embedded Computing Systems (TECS) 20(1):1–18
24. Stellari F, Song P, Weger AJ, Culp J, Herbert A, Pfeiffer D (2014) Verification of untrusted chips using trusted layout and emission measurements. In: Hardware-Oriented Security and Trust (HOST), 2014 IEEE International Symposium On, pp 19–24. IEEE
25. Haut JM (2017) Paoletti M, Plaza J, Plaza A Cloud implementation of the k-means algorithm for hyperspectral image analysis. J Supercomput 73:514–529
26. Atlam H (2020) Developing Dynamic and Adaptive Risk-based Access Control Model for The Internet of Things. Doctoral dissertation, University of Southampton. https://eprints.soton.ac.uk/447742/1/Final_Thesis_Hany_Atlam.pdf
27. Gong X, Richman MB (1995) On the application of cluster analysis to growing season precipitation data in north america east of the rockies. J Clim 8(4):897–931

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.